

Data generation module

S175 WeatherRouting performance simulator

Full factorial design of experiments
(grid spacing: $\Delta H_s = 0.25$ m, $\Delta \chi = 5^\circ$)
126,153,720 cases;
infeasible outputs encoded as -1

Screening campaign

Stratified 5% subsample
screening-training / validation /
held-out selection

Final training campaign

Independent stratified 80/10/10 split
input and regression-target standardisation
using final-training statistics

Structured holdout campaigns

Level, block, and corner holdouts
complete two-stage surrogate retrained per regime
each regime has its own validation partition

Algorithm and loss-weight screening module (5% stratified subsample)

*Model-family screening target:
ship speed; infeasibility encoded as -1*

Random Forest

XGBoost

MLP
(256-128-64)

Compare on identical screening partitions
MLP family selected for C1, C2, and R;
R extended to multi-output prediction

Loss-weight screening

*Ten-output MLP (512-256-128) trained on fully valid and
fuel-only-infeasible screening cases; fuel target masked where
undefined;
selection metrics evaluated on the held-out selection partition*

Joint loss-weight grid

$\alpha_{\text{speed}} \in \{1.0, 1.5, 2.0, 3.0, 5.0\}$
 $\alpha_{\text{others}} \in \{1.5, 1.0, 0.8, 0.67, 0.5, 0.33\}$
30 pairs $\times$ 5 seeds, trained jointly using the final-regressor protocol

Prespecified selection rule

- Eligibility in every seed, evaluated on positive-target selection rows:
ship speed under-predicted in more than 50%; each of the other nine
outputs over-predicted in more than 50%.
- Among eligible pairs: lowest mean standardised MAE; exact ties
broken by higher mean R^2 .

Selected for the final campaign

MLP family for C1, C2, and R Only eligible pair among the 30 tested:

- $(\alpha_{\text{speed}}, \alpha_{\text{others}}) = (1.5, 0.33)$
- $\alpha_{\text{other}} = 0.33$ is the lower boundary of the evaluated grid.

Two-stage surrogate construction module (final training campaign)

Label final-training cases:
fully valid / fuel-only infeasible / completely infeasible

Stage 1: Feasibility classification

Classifier C1: MLP (512-256-128)
complete infeasibility
trained on all final-training cases

Classifier C2: MLP (512-256-128-64)
fuel-only infeasibility
trained on cases that are not
completely infeasible

Thresholds τ_1 and τ_2 selected on final validation:
highest thresholds maintaining
infeasible-class recall ≥ 0.99

Stage 2: Multi-output regression

Final multi-output MLP regressor (512-256-128)
fully valid + fuel-only-infeasible training cases
fuel target masked where undefined
selected weights: $\alpha_{\text{speed}} = 1.5$; $\alpha_{\text{others}} = 0.33$

*Residuals empirically lie preferentially on the prescribed conservative
side for all ten outputs.*

Assembled two-stage surrogate
 $C1 \geq \tau_1 \rightarrow$ all outputs = -1; else regressor R;
 $C2 \geq \tau_2 \rightarrow$ fuel = -1

Evaluation
primary screening-disjoint test subset (11,984,559 rows)
screening-disjoint structured holdouts
fresh off-grid simulator runs
interpolation baseline and loss ablation